

Hamming Code

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Hamming

❖ Basics of Hamming Code

- ❑ It is given by RW Hamming
- ❑ It is used to detect and correct errors.
- ❑ In Hamming Code, Codeword = Data bits + Parity bits
- ❑ It is represented by (n,k) code
- ❑ Where, n = Total bits and k = message bits
- ❑ Parity bits $p = n - k$

❖ Parity bits condition in Hamming Code

❑ To identify Parity bits, it should satisfy the given condition.

$$\leadsto \text{message bits } k = 4 \quad \therefore 2^p \geq p + k + 1$$

$$\Rightarrow 2^p \geq p + 4 + 1$$

$$\Rightarrow 2^p \geq p + 5$$

$$\leadsto \text{If } p = 1$$
$$\Rightarrow 2^1 \geq 6$$

x

$$\leadsto p = 2$$
$$\Rightarrow 2^2 \geq 7$$

x

$$\leadsto p = 3$$
$$\Rightarrow 2^3 \geq 8$$

✓

$\leadsto$ Minimum value of $p = 3$ bits.

$\leadsto$ Hamming Code $(n, k) = (\underline{7}, 4)$

❖ Position of Parity bits & Message bits in Hamming Code

□ Let's take an example of (7,4) hamming code.

$$P_1 = 2^0 = 1$$

$$P_2 = 2^1 = 2$$

$$P_4 = 2^2 = 4$$

Bit

Positions

5

4

3

2

1

7 D_7	D_6	D_5	P_4	D_3	P_2	P_1
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6

❖ Calculation of Parity bits in Hamming Code

□ Let's take an example of (7,4) hamming code.

Bit

Positions

5

4

3

2

1

7						
D7	D6	D5	P4	D3	P2	P1

6

$$\begin{aligned} \Rightarrow P_1 &= D_3 \oplus D_5 \oplus D_7 \\ P_2 &= D_3 \oplus D_6 \oplus D_7 \\ P_4 &= D_5 \oplus D_6 \oplus D_7 \end{aligned} \quad \left. \vphantom{\begin{aligned} \Rightarrow P_1 &= D_3 \oplus D_5 \oplus D_7 \\ P_2 &= D_3 \oplus D_6 \oplus D_7 \\ P_4 &= D_5 \oplus D_6 \oplus D_7 \end{aligned}} \right\} \text{even parity}$$

❖ Generation of Hamming Code

□ If 5 bits data 01101 is given. Represent given data in hamming code.

⇒ $k = 5$ bits, data = 01101

⇒ Cond.ⁿ of parity bits

$$\Rightarrow 2^p \geq p + k + 1$$

$$\Rightarrow 2^p \geq p + 6$$

⇒ $p = 1$

$$\Rightarrow 2^1 \geq 7$$

X

⇒ $p = 2$

$$\Rightarrow 2^2 \geq 8$$

X

⇒ $p = 3$

$$\Rightarrow 2^3 \geq 9$$

X

⇒ $p = 4$

$$\Rightarrow 2^4 \geq 10$$

✓

⇒ $p = 4$ bits

$$p_1 = 2^0 = 1$$

$$p_2 = 2^1 = 2$$

$$p_4 = 2^2 = 4$$

$$p_8 = 2^3 = 8$$

Bit Positions		7	6	5	4	3	2	1
D₉	P₈	D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁
0	0	1	1	0	0	1	1	0

$$\Rightarrow P_1 = 1 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$P_2 = 1 \oplus 1 \oplus 1 = 1$$

$$P_4 = 1 \oplus 1 \oplus 0 = 0$$

$$P_8 = 0$$

$$\Rightarrow \text{Codeword} = \underline{001100110}$$

↓
x

❖ Error Detection and Correction by Hamming Code

❑ If the received code as per Hamming code is 001101110. then find error in received codeword.

Bit Positions								
9	0	1	1	0	1	1	1	0
D_9	P_8	D_7	P_6	D_5	P_4	D_3	P_2	P_1

$$\rightarrow P_1 = 1 \oplus 0 \oplus 1 \oplus 0 = 0 \quad \checkmark$$

$$P_2 = 1 \oplus 1 \oplus 1 = 1 \quad \checkmark$$

$$P_4 = 1 \oplus 1 \oplus 0 = 0 \quad \times$$

$$P_8 = 0 = 0 \quad \checkmark$$

$$\rightarrow {}_5P_8 P_4 P_2 P_1 = \checkmark \times \checkmark \checkmark = 0100 = (4)$$

Error
Syndrome

$\oplus$

Received
data

=

Corrected
data

9	8	7	6	5	4	3	2	1
0	0	0	0	0	1	0	0	0
9	8	7	6	5	4	3	2	1
0	0	1	1	0	1	1	1	0
9	8	7	6	5	4	3	2	1
0	0	1	1	0	0	1	1	0